

WAZUH Home Lab - SIEM and File Integrity Monitoring

By VIKNESH V S

Table of Contents

1. Overview
2. Lab Architecture
3. Prerequisites
4. Installing the Wazuh Manager (Ubuntu)
 - 4.1 Add Wazuh GPG Key
 - 4.2 Download and Execute Wazuh Installation Script
5. Accessing the Wazuh Dashboard
6. Registering the Agent with the Manager
7. Generate Agent Key on Ubuntu Manager
 - 7.1 Apply Key in the Windows Agent
8. File Integrity Monitoring on Windows
 - 8.1 Edit Agent Configuration
 - 8.2 Restart the Agent
9. Verifying Setup

1. Overview

Wazuh is a free, open-source security platform that offers:

- Log analysis
- File integrity monitoring
- Intrusion detection
- Vulnerability detection
- Real-time alerting

2. Lab Architecture

Component	Host	Role
Wazuh Manager	Ubuntu (VirtualBox)	Collects, analyzes, and stores data from agents
Wazuh Agent	Windows (host machine)	Sends logs and system events to the Wazuh manager

Network Configuration: Use Bridged Adapter in VirtualBox to place the Ubuntu server on the same network as the host. This allows access between host and guest.

3. Prerequisites

- VirtualBox installed
- Ubuntu Server 20.04+ installed in VirtualBox (bridged networking)
- Internet access on Ubuntu VM
- Administrative access on the Windows host
- Optional: basic knowledge of Linux and system administration

4. Installing the Wazuh Manager (Ubuntu)

Run the following steps on your Ubuntu VirtualBox server

4.1 Add Wazuh GPG Key

Via a terminal window

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH |  
sudo gpg --dearmor -o /usr/share/keyrings/wazuh-archive-  
keyring.gpg
```

This adds the GPG key to verify Wazuh packages.

4.2 Download and Execute Wazuh Installation Script

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh &&  
sudo bash ./wazuh-install.sh -a -i
```

- -a: Installs all components (manager, indexer.)
- -i: Runs in interactive mode

The script installs all required services and configures them automatically.

5. Accessing the Wazuh Dashboard

After installation:

Check your Ubuntu VM's IP address:

`ifconfig`

On your Ubuntu server, open a browser and go to:

[https:// <ubuntu-vm-ip>](https://<ubuntu-vm-ip>)

1. Accept any browser security warning due to the self-signed certificate.
2. Log in using the credentials displayed at the end of the installation script.

6. Installing the Wazuh Agent (Windows Host)

Download the latest Wazuh agent MSI installer from the official documentation:

<https://documentation.wazuh.com/current/installation-guide/wazuh-agent/wazuh-agent-package-windows.html>

Install the MSI package on your Windows system using the default settings

7. Registering the Agent with the Manager

7.1 Generate Agent Key on Ubuntu Manager

Run the agent management utility:

```
sudo /var/ossec/bin/manage_agents
```

- Select A to add an agent.
- Assign a name (e.g., WindowsHost).
- Leave IP address blank unless static assignment is needed.
- After creation, select E to extract the key.
- Copy the key output.

7.2 Apply Key in the Windows Agent

1. Open Wazuh Agent Manager GUI from the Start Menu.
2. Paste the copied key into the appropriate field.
3. Save and apply the key.
4. Add the manager's IP address (IP address of your Ubuntu manager).
5. Restart the agent service. You can then go to the WAZUH dashboard and see the agent onboarded.

8. File Integrity Monitoring on Windows

Wazuh supports real-time monitoring of file and folder changes using Syscheck

8.1 Edit Agent Configuration

Open the following configuration file:

[C:\Program Files \(x86\)\ossec-agent\ossec.conf](#)

Add the following entry inside the Directory block:

```
<directories realtime="yes" >C:\Users\abc\Test</directories>
```

This monitors the specified folder in real-time.

8.2 Restart the Agent

After saving the changes, restart the Wazuh agent service to apply the configuration.

9. Verifying Setup

1. Open the Wazuh Dashboard in your browser.
2. Navigate to Agents → ensure the Windows agent is listed and status is Active.
3. Go to the Integrity Monitoring section.
4. Perform actions (create/modify/delete files) in the monitored folder.
5. Confirm that alerts appear in the dashboard.